

Shape weights on ribbons: the commuting locus is the transpose-equivariant $\gcd(e, f)$ -periodic characters

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Abstract

Let R_e^W add an e -ribbon to a partition with a weight W depending on the *whole shape* of the ribbon — 2^{e-1} free parameters, one per composition of e , rather than the e parameters of a height-only weight. We determine, for $1 \leq e < f$ over an integral domain, every pair $(W, \bar{W})$ with $[R_e^W, R_f^{\bar{W}}] = 0$. The answer is not the Murnaghan–Nakayama anchor alone. Writing $d = \gcd(e, f)$, the commuting locus consists of the weights

$$W(u) = W(0^{e-1}) \prod_{j=1}^{e-1} (-\gamma(j \bmod d))^{u_j},$$

where $u \in \{0, 1\}^{e-1}$ is the occupancy word of the ribbon and $\gamma : \mathbb{Z}/d \rightarrow F^\times$ is any function with $\gamma(0) = 1$ and $\gamma(\delta)\gamma(-\delta) = 1$; and $\bar{W}$ is given by the same γ . This is a torus of dimension $\lfloor (d-1)/2 \rfloor$ together with a $\mathbb{Z}/2$ when d is even. For $\gcd(e, f) = 1$ it collapses to the Murnaghan–Nakayama sign, so the Q147 theorem extends verbatim from height-only to shape weights; but for $\gcd(e, f) > 1$ the locus is strictly larger, and for $\gcd(e, f) \geq 3$ it is positive-dimensional.

The two sectors of the commutator cut out the two halves of the answer, and each half has a name. The **one-bead sector** forces W to be *multiplicative* in the letters of the occupancy word and *d-periodic* in their positions. The **two-bead sector** forces W to be *equivariant under ribbon transposition*. The Murnaghan–Nakayama sign is the unique such weight when $\gcd(e, f) = 1$; in general it is the unique one *up to a diagonal gauge*, and we exhibit the gauge explicitly on a subtorus and verify it computationally on the whole locus.

Everything below is derived from the operator: the weight enters only as an opaque symbol indexed by a word, and no step uses $(-1)^{\text{ht}}$, t^{ht} , or height-only dependence as an input.

Contents

1	The question, and why the shape version is not exotic	1
2	The sign is a gauge: σ-coordinates	3
3	Realisability: every local pattern occurs	3
4	The matrix elements	3
4.1	The one-bead sector	4
4.2	The two-bead sector	4
5	The answer	5

6	The cheapest decisive test, by hand	6
7	Proof of Theorem 5.2: sufficiency	6
8	Proof of Theorem 5.2: necessity	7
8.1	Flip relations (polynomial form)	7
8.2	Nonvanishing	7
8.3	Constancy of the flip ratios	8
8.4	Identification of γ	9
9	What the extra freedom is: the d -quotient, and gauge	9
9.1	The special points: Murnaghan–Nakayama one level down	9
9.2	The torus directions are a diagonal gauge	10
9.3	Transpose-equivariance is exactly the two-bead condition	10
10	Computational verification	11
11	What is and is not proved	12

1 The question, and why the shape version is not exotic

Maya sets and the ribbon operator

Let λ be a partition and $M = M(\lambda) = \{\lambda_j - j : j \geq 1\}$ its Maya set (beta-set): a subset of $\mathbb{Z}$ containing all sufficiently negative integers and bounded above. Adding a connected e -ribbon to λ is the same as a *bead move* $b \mapsto b + e$ with $b \in M$ and $b + e \notin M$.

Convention 1.1 (occupancy word). For a legal move $b \mapsto b + e$ put

$$u_M(b, e) = (u_1, \dots, u_{e-1}) \in \{0, 1\}^{e-1}, \quad u_i = [b + i \in M].$$

We call u the *occupancy word* of the move. Its weight $|u| = \sum u_i$ is the number of beads jumped, i.e. the height ht of the ribbon in the usual normalisation $\text{ht} = \#\text{rows} - 1$.

Convention 1.2 (the operator). A *shape weight* of rank e over a commutative ring K is a function $W : \{0, 1\}^{e-1} \rightarrow K$. It defines a K -linear operator on the span of the Maya sets,

$$R_e^W |M\rangle = \sum_{\substack{b \in M \\ b+e \notin M}} W(u_M(b, e)) |(M \setminus \{b\}) \cup \{b+e\}\rangle,$$

equivalently on symmetric functions, $R_e^W s_\lambda = \sum_\mu W(u) s_\mu$ over the e -ribbons μ/λ . A *height weight* is the special case $W(u) = w_{|u|}$.

The dictionary: words are ribbon shapes

Lemma 1.3 (word $\leftrightarrow$ composition). *Let $u \in \{0, 1\}^{e-1}$ and let $0 = c_0 < c_1 < \dots < c_k < c_{k+1} = e$ list $\{0\} \cup \{i : u_i = 1\} \cup \{e\}$. Then the row lengths of the ribbon μ/λ produced by the move, read **top row first**, are*

$$\alpha(u) = (c_{k+1} - c_k, c_k - c_{k-1}, \dots, c_1 - c_0) \models e.$$

The map $u \mapsto \alpha(u)$ is a bijection $\{0, 1\}^{e-1} \rightarrow \{\text{compositions of } e\}$ with $|u| = \ell(\alpha) - 1 = \text{ht}$.

Proof. The gaps between consecutive occupied sites of $\{b\} \cup (M \cap (b, b+e)) \cup \{b+e\}$ are the row lengths, because in beta-coordinates a *larger* bead sits in an *earlier* (higher) row; hence the reversal. Verified against an independent brute-force enumeration over Young diagrams on 1375 ribbons, $|\lambda| \leq 8$, $e \leq 5$ (§10, Check V1). $\square$

Remark 1.4 (the reversal is not cosmetic). The brief for this session recorded the dictionary without the reversal. The cross-check of Lemma 1.3 against the Young-diagram enumeration failed on the very first case ($\lambda = \emptyset$, $e = 3$, $\mu = (2, 1)$) and located it. With the reversal dropped the two L-trominoes $(2, 1)$ and $(1, 2)$ exchange names, which is precisely the distinction the cheapest test of §6 is about.

So the question under test is an honest one about ribbon shapes:

Question 1.5. For which pairs of shape weights $(W, \bar{W})$ of ranks $e < f$ over an integral domain K is $[R_e^W, R_f^{\bar{W}}] = 0$?

Height weights are the subfamily $W(u) = w_{|u|}$. The theorem of [1, 2026-09-11 c2] ("free height weights force Murnaghan–Nakayama") is the restriction of Question 1.5 to that subfamily. The hypothesis under test today is therefore not a parameter but a *form*: does the weight need to see only how many rows the ribbon has?

Ribbon transposition, correctly

Lemma 1.6 (transposition on words). *Transposing a ribbon acts on occupancy words by **reversal followed by complementation**:*

$$(u^T)_i = 1 - u_{e-i}, \quad 1 \leq i \leq e-1.$$

In particular $|u| + |u^T| = e-1$, recovering $\text{ht}(R) + \text{ht}(R^T) = e-1$.

Proof. Direct verification against conjugation of Young diagrams, 1500 ribbons, $e \leq 7$, $|\lambda| \leq 7$ (§10, Check V2). The identity $|u| + |u^T| = e-1$ is immediate from the formula. $\square$

Remark 1.7 (a correction to the brief). The brief asserted that transposition is *reversal* on compositions, and drew two conclusions from it: that the induced $\mathbb{Z}/2$ fixes the height-only subfamily pointwise, and that it exchanges the two L-trominoes $(2, 1) \leftrightarrow (1, 2)$. Both are false. Reversal alone agrees with transposition on only 75 of 765 tested ribbons. By Lemma 1.6, for $e = 3$ transposition *fixes* both L-trominoes and exchanges $(3) \leftrightarrow (1, 1, 1)$; and on height weights it acts by $w_N \mapsto w_{e-1-N}$, which is not the identity. The correction matters, because transpose-equivariance turns out to be exactly the content of the two-bead sector (Theorem 5.2 (iii)).

2 The sign is a gauge: σ -coordinates

Convention 2.1. For a shape weight W of rank e set

$$\sigma(u) := (-1)^{|u|} W(u), \quad u \in \{0, 1\}^{e-1}.$$

Murnaghan–Nakayama is $\sigma \equiv \text{const.}$

The whole point of this substitution is Lemmas 4.2 and 4.3: in σ -coordinates *every* matrix element of the commutator is a difference of two products with $+$ signs. The alternating sign of Murnaghan–Nakayama is not a feature of the commuting locus; it is a change of coordinates that the commutator cannot see. Everything interesting happens after it is removed.

3 Realisability: every local pattern occurs

Lemma 3.1. *Let $g \geq 2$ and $S \subseteq \{1, \dots, g-1\}$. There is a partition λ with $0 \in M(\lambda)$, $g \notin M(\lambda)$ and $M(\lambda) \cap (0, g) = S$.*

Proof. Put $r = 1 + |S|$ and $M = \mathbb{Z}_{\leq -r-1} \cup \{0\} \cup S$. Then M contains all sufficiently negative integers, is bounded above, and has charge $\#(M \cap \mathbb{Z}_{\geq 0}) - \#(\mathbb{Z}_{<0} \setminus M) = (1 + |S|) - r = 0$; so $M = M(\lambda)$ for a unique λ . The r deleted sites lie below 0 and miss the window. $\square$

Corollary 3.2. *Every word $u \in \{0, 1\}^{e-1}$ occurs as $u_M(b, e)$ for some partition. More generally every window configuration used below may be prescribed freely.*

Corollary 3.2 is the control that stops a free parameter from being free for a trivial reason. It is confirmed computationally for $e \leq 6$: all 2^{e-1} words occur already among partitions of size ≤ 14 (§10, Check C2).

4 The matrix elements

Fix $1 \leq e < f$ and put $g = e + f$. By Lemma 4.1 below, a commutator matrix element $\langle s_\mu, [R_e^W, R_f^{\bar{W}}] s_\lambda \rangle$ can be nonzero only when $|M(\lambda) \triangle M(\mu)| \in \{2, 4\}$.

Lemma 4.1 (sectors). $\langle s_\mu, [R_e^W, R_f^{\bar{W}}] s_\lambda \rangle = 0$ unless $|M(\lambda) \triangle M(\mu)| \in \{2, 4\}$.

Proof. Each $R_\bullet^{\bullet}$ moves one bead, so a product moves at most two; and $M(\mu) = M(\lambda)$ is impossible since $|\mu| = |\lambda| + g > |\lambda|$. (Unchanged from [1]: the argument never looks at the weight.) $\square$

4.1 The one-bead sector

Lemma 4.2 (one-bead element, shape form). *Let $b \in M$, $b+g \notin M$ and $M(\mu) = (M \setminus \{b\}) \cup \{b+g\}$. Write the window word*

$$X = M|_{(b, b+g)} = A m_e V m_f D \in \{0, 1\}^{g-1},$$

so that $A = X_{[1, e-1]}$, $m_e = X_e$, $V = X_{[e+1, f-1]}$, $m_f = X_f$, $D = X_{[f+1, g-1]}$; and put $C = A m_e V = X_{[1, f-1]}$, $B = V m_f D = X_{[e+1, g-1]}$. Then

$$\langle s_\mu, [R_e^W, R_f^{\bar{W}}] s_\lambda \rangle = (1 - 2m_f) W(D) \bar{W}(C) - (1 - 2m_e) W(A) \bar{W}(B),$$

and in σ -coordinates

$$\boxed{\sigma(D) \bar{\sigma}(C) - \sigma(A) \bar{\sigma}(B)} \tag{1}$$

up to the global factor $(-1)^{|X| - m_e - m_f}$. By Lemma 3.1, $A, D \in \{0, 1\}^{e-1}$, $V \in \{0, 1\}^{f-e-1}$ and $m_e, m_f \in \{0, 1\}$ may be prescribed independently.

Proof. R_e^W and $R_f^{\bar{W}}$ act by legal bead moves. Exactly as in [1]: since $|M \triangle M(\mu)| = 2$ while two independent moves would give 4, in each of the two products one added site must cancel one removed site, which forces the two routes

$$(B) \quad M \xrightarrow{(b, e)} M_1 \xrightarrow{(b+e, f)} M(\mu), \quad (A) \quad M \xrightarrow{(b+f, e)} M'_1 \xrightarrow{(b, f)} M(\mu).$$

Route (B) is legal iff $b + e \notin M$, i.e. $m_e = 0$; route (A) is legal iff $b + f \in M$, i.e. $m_f = 1$. Their occupancy words: in route (B) the first move sees $M \cap (b, b+e) = A$ and the second sees

$M_1 \cap (b+e, b+g)$; M_1 differs from M only at b and $b+e$, both outside the *open* interval $(b+e, b+g)$, so this is B . Symmetrically route (A) gives D then C . Hence

$$\langle s_\mu, R_f^{\bar{W}} R_e^W s_\lambda \rangle = [m_e = 0] W(A) \bar{W}(B) + [m_f = 1] W(D) \bar{W}(C),$$

$$\langle s_\mu, R_e^W R_f^{\bar{W}} s_\lambda \rangle = [m_f = 0] \bar{W}(C) W(D) + [m_e = 1] \bar{W}(B) W(A),$$

and the difference is $(1 - 2m_f)W(D)\bar{W}(C) - (1 - 2m_e)W(A)\bar{W}(B)$.

For the σ -form: $X = A m_e V m_f D$ gives $|C| + |D| = |X| - m_f$ and $|A| + |B| = |X| - m_e$, so

$$(1 - 2m_f)W(D)\bar{W}(C) = (1 - 2m_f)(-1)^{|X| - m_f} \sigma(D) \bar{\sigma}(C) = (-1)^{|X|} \sigma(D) \bar{\sigma}(C),$$

using $(1 - 2m)(-1)^{-m} = 1$ for $m \in \{0, 1\}$; likewise for the other term. $\square$

4.2 The two-bead sector

Lemma 4.3 (two-bead element, shape form). *Let $2 \leq e < f$, $1 \leq \delta \leq e - 1$, and take $b = 0$, $c = \delta$ with $0, \delta \in M$ and $e, \delta + f \notin M$, $M(\mu) = (M \setminus \{0, \delta\}) \cup \{e, \delta + f\}$. Write*

$$\pi = M|_{(0, \delta)} \in \{0, 1\}^{\delta-1}, \quad \pi' = M|_{(\delta, e)} \in \{0, 1\}^{e-\delta-1}, \quad \theta' = M|_{(e, \delta+f)} \in \{0, 1\}^{\delta+f-e-1},$$

all prescribable freely. Then in σ -coordinates the matrix element is, up to an overall sign,

$$\boxed{\sigma(\pi 0 \pi') \bar{\sigma}(\pi' 0 \theta') - \sigma(\pi 1 \pi') \bar{\sigma}(\pi' 1 \theta')} \quad (2)$$

where in $\sigma(\pi b \pi')$ the distinguished letter sits at position δ , and in $\bar{\sigma}(\pi' b \theta')$ at position $e - \delta$.

Proof. The four sites $0, \delta, e, \delta + f$ are pairwise distinct ($1 \leq \delta \leq e - 1 < f$). Performing the e -move first gives weights $W(\Pi), \bar{W}(\Theta')$ with $\Pi = M|_{(0, e)}$ and $\Theta' = M_1|_{(\delta, \delta+f)}$, $M_1 = (M \setminus \{0\}) \cup \{e\}$; performing the f -move first gives $\bar{W}(\Theta), W(\Pi')$ with $\Theta = M|_{(\delta, \delta+f)}$ and $\Pi' = M_2|_{(0, e)}$, $M_2 = (M \setminus \{\delta\}) \cup \{\delta + f\}$. Every other route would move a different pair of beads, and the pair is determined because $e \neq f$ ([2, Thm. 5(2)]). Now $\Pi = \pi 1 \pi'$ (the letter at δ is 1 since $\delta \in M$) and $\Pi' = \pi 0 \pi'$ (the site $\delta + f$ lies outside $(0, e)$); while $\Theta = \pi' 0 \theta'$ (the letter at e , i.e. index $e - \delta$, is 0 since $e \notin M$) and $\Theta' = \pi' 1 \theta'$ (the site 0 lies outside $(\delta, \delta + f)$). So the element is $\bar{W}(\pi' 0 \theta') W(\pi 0 \pi') - W(\pi 1 \pi') \bar{W}(\pi' 1 \theta')$, and since $|\pi 0 \pi'| + |\pi' 0 \theta'|$ and $|\pi 1 \pi'| + |\pi' 1 \theta'|$ differ by 2, the two products acquire the *same* sign under Convention 2.1. Freeness of π, π', θ' is Lemma 3.1 applied with $g = \delta + f$. $\square$

Remark 4.4. The two-bead sector is empty when $\min(e, f) = 1$ ([2, Cor. 9]), so for $e = 1$ the answer must come from the one-bead sector alone. It does; see Proposition 8.5.

5 The answer

Convention 5.1. Let $d \mid e$ and let $\gamma : \mathbb{Z}/d \rightarrow F^\times$ satisfy $\gamma(0) = 1$. Write

$$W_e^\gamma(u) := \prod_{j=1}^{e-1} (-\gamma(j \bmod d))^{u_j}, \quad \text{equivalently} \quad \sigma_e^\gamma(u) = \prod_{j=1}^{e-1} \gamma(j \bmod d)^{u_j}.$$

Theorem 5.2 (classification). *Let $1 \leq e < f$, $d = \gcd(e, f)$, let K be an integral domain with fraction field F , and let $W, \bar{W}$ be shape weights of ranks e, f over K . Then $[R_e^W, R_f^{\bar{W}}] = 0$ if and only if one of*

(a) $W \equiv 0$; (b) $\bar{W} \equiv 0$;

(c) there are $\alpha, \beta \in K \setminus \{0\}$ and a function $\gamma : \mathbb{Z}/d \rightarrow F^\times$ with

$$\gamma(0) = 1 \quad \text{and} \quad \gamma(\delta) \gamma(-\delta) = 1 \quad (\delta \in \mathbb{Z}/d),$$

such that $W = \alpha W_e^\gamma$ and $\bar{W} = \beta W_f^\gamma$.

Moreover in case (c) the three conditions on $(W, \bar{W})$ are exactly:

- (i) multiplicativity: $W(u) = W(0^{e-1}) \prod_j c_j^{u_j}$ for scalars c_j ;
- (ii) d -periodicity: c_j depends only on $j \bmod d$, and $c_j = 1$ when $d \mid j$; the same c serves for W and for $\bar{W}$;
- (iii) transpose-equivariance: $W \circ T = \kappa W$ and $\bar{W} \circ T = \bar{\kappa} \bar{W}$ for scalars $\kappa, \bar{\kappa}$, with T as in Lemma 1.6.

The one-bead sector is equivalent to (i)+(ii); the two-bead sector, given (i)+(ii), is equivalent to (iii).

Before the proof, the consequences.

Corollary 5.3 ($\gcd(e, f) = 1$: the Q147 theorem survives the enlargement). *If $\gcd(e, f) = 1$ then γ is the trivial function on $\mathbb{Z}/1$, so (c) reads*

$$W(u) = \alpha (-1)^{|u|}, \quad \bar{W}(y) = \beta (-1)^{|y|},$$

i.e. $R_e^W = \alpha p_e \cdot$ and $R_f^{\bar{W}} = \beta p_f \cdot$. Shape dependence is forced away: the 2^{e-1} parameters collapse to one.

Corollary 5.4 (the size of the locus). *The set of γ in (c) is a group isomorphic to $(F^\times)^{\lfloor (d-1)/2 \rfloor} \times \mu_2(F)^{[2|d]}$. In particular the locus is a single point (modulo the scalars α, β) iff $d = 1$; it is finite of size 2 iff $d = 2$; and it is positive-dimensional, of dimension $\lfloor (d-1)/2 \rfloor \geq 1$, as soon as $d \geq 3$.*

Proof. γ is free on a set of representatives of the orbits $\{\delta, -\delta\}$ with $\delta \neq -\delta$, of which there are $(d - \gcd(2, d))/2 = \lfloor (d-1)/2 \rfloor$; on the fixed points $\delta = -\delta$ the condition reads $\gamma(\delta)^2 = 1$, and $\gamma(0) = 1$ is imposed. $\square$

Corollary 5.5 (the cheapest decisive test). *For $(e, f) = (1, 3)$: $d = 1$, so the commutator forces the two **L-trominoes** equal, $\bar{W}_{(2,1)} = \bar{W}_{(1,2)}$, and then forces the full alternation $\bar{W}_\alpha = \beta (-1)^{\ell(\alpha)-1}$.*

6 The cheapest decisive test, by hand

It is worth seeing the mechanism once with no machinery. Take $e = 1, f = 3$. Then $A = D = \emptyset$, W is the single scalar $\sigma(\emptyset) =: \alpha$, and $V \in \{0, 1\}$. Lemma 4.2 gives, for all m_e, m_f, V ,

$$\alpha \bar{\sigma}(m_e V) = \alpha \bar{\sigma}(V m_f).$$

Assume $\alpha \neq 0$. The four choices of (m_e, m_f) with $V \in \{0, 1\}$ give

$$\bar{\sigma}_{01} = \bar{\sigma}_{10}, \quad \bar{\sigma}_{00} = \bar{\sigma}_{00}, \quad \bar{\sigma}_{10} = \bar{\sigma}_{01}, \quad \bar{\sigma}_{11} = \bar{\sigma}_{11}, \quad \bar{\sigma}_{0V} = \bar{\sigma}_{V0}, \quad \bar{\sigma}_{1V} = \bar{\sigma}_{V1},$$

i.e. $\bar{\sigma}$ is unchanged by the cyclic shift $y_1 y_2 \mapsto y_2 y_1$ and by changing the first letter; iterating, $\bar{\sigma}$ is constant. Since $\bar{\sigma}_{10}$ is the word $(1, 0) \leftrightarrow \alpha = (2, 1)$ and $\bar{\sigma}_{01}$ is $(0, 1) \leftrightarrow (1, 2)$, the equation $\bar{\sigma}_{01} = \bar{\sigma}_{10}$ is the statement that the cross-rank commutator cannot distinguish the two L-trominoes — and it does not merely fail to distinguish them, it forbids any weight that does. Undoing Convention 2.1, $\bar{W}(y) = \beta(-1)^{|y|}$, the Murnaghan–Nakayama sign.

The raw operator produces exactly five distinct nonzero equations here, all of the form $\alpha \cdot (\bar{W}_y \pm \bar{W}_{y'})$; see §10, Check E1.

7 Proof of Theorem 5.2: sufficiency

Proposition 7.1. *Let $d \mid \gcd(e, f)$ and let $\gamma : \mathbb{Z}/d \rightarrow F^\times$ satisfy $\gamma(0) = 1$ and $\gamma(\delta)\gamma(-\delta) = 1$. Then $[R_e^{W_e^\gamma}, R_f^{W_f^\gamma}] = 0$.*

Proof. Write $\Sigma_M(x, y) = \prod_{i \in M \cap (x, y)} \gamma((i - x) \bmod d)$ for $d \mid (y - x)$, so that $\sigma^\gamma(u_M(b, e)) = \Sigma_M(b, b + e)$.

One-bead sector. With the notation of Lemma 4.2, index the window $(b, b + g)$ by $i = 1, \dots, g - 1$ and put $P = \prod_{i \in M \cap (b, b + g)} \gamma(i \bmod d)$. Since $d \mid e$ and $d \mid f$,

$$\sigma(A)\bar{\sigma}(B) = \Sigma_M(b, b + e) \Sigma_M(b + e, b + g) = P \cdot \gamma(e \bmod d)^{-m_e} = P,$$

because the two factors together run over all occupied i in the window except $i = e$, and $\gamma(e \bmod d) = \gamma(0) = 1$; the shift of the second interval's base point by e is invisible to γ because $d \mid e$. Identically $\sigma(D)\bar{\sigma}(C) = P \cdot \gamma(f \bmod d)^{-m_f} = P$. So (1) vanishes.

Two-bead sector. With the notation of Lemma 4.3,

$$\frac{\sigma(\pi 1 \pi')}{\sigma(\pi 0 \pi')} = \gamma(\delta \bmod d), \quad \frac{\bar{\sigma}(\pi' 1 \theta')}{\bar{\sigma}(\pi' 0 \theta')} = \gamma((e - \delta) \bmod d) = \gamma(-\delta \bmod d),$$

the last equality because $d \mid e$. Their product is 1 by hypothesis, so (2) vanishes. By Lemma 4.1 these are all the matrix elements. $\square$

Remark 7.2. The sufficiency computation is the Q147 one with the count N of *all* jumped beads replaced by the γ -weighted count, and the two-bead computation is where $\gamma(\delta)\gamma(-\delta) = 1$ enters — once, and decisively. Note that the one-bead sector imposes *no* condition on γ beyond $\gamma(0) = 1$.

8 Proof of Theorem 5.2: necessity

Throughout, K is a domain with fraction field F , $[R_e^W, R_f^{\bar{W}}] = 0$, $\sigma \not\equiv 0$ and $\bar{\sigma} \not\equiv 0$. Write $a = e - 1$, $q = f - e$, $n = f - 1$. For a word z and an index j , $z^{(j)}$ denotes z with letter j flipped.

8.1 Flip relations (polynomial form)

Lemma 8.1. *The following hold for all admissible arguments.*

(F0) $\bar{\sigma}(y)$ does not depend on y_e nor on y_{f-e} .

(F1) For $1 \leq j \leq e - 1$: $\sigma(A^{(j)})\bar{\sigma}(A m V) = \sigma(A)\bar{\sigma}(A^{(j)} m V)$.

(F2) For $1 \leq i \leq f - e - 1$: $\bar{\sigma}(A m V)\bar{\sigma}(V^{(i)} m' D) = \bar{\sigma}(A m V^{(i)})\bar{\sigma}(V m' D)$.

(F3) For $1 \leq i \leq e-1$: $\sigma(D^{(i)}) \bar{\sigma}(V m' D) = \sigma(D) \bar{\sigma}(V m' D^{(i)})$.

Proof. Write (1) as $\sigma(D) \bar{\sigma}(A m_e V) = \sigma(A) \bar{\sigma}(V m_f D)$, valid for all independent A, D, V, m_e, m_f (Lemma 3.1).

(F0) Flipping m_e changes the left side only: $\sigma(D) [\bar{\sigma}(A 0 V) - \bar{\sigma}(A 1 V)] = 0$. Choose D with $\sigma(D) \neq 0$ and use that K is a domain. The letter m_e occupies position e of $C = A m_e V$, and C ranges over all of $\{0, 1\}^{f-1}$. Flipping m_f gives the statement for position $f-e$ of $B = V m_f D$.

(F1) Flipping A_j gives $\sigma(D) \bar{\sigma}(A^{(j)} m_e V) = \sigma(A^{(j)}) \bar{\sigma}(V m_f D)$. Multiply (1) by $\sigma(A^{(j)})$ and this by $\sigma(A)$ and subtract: $\sigma(D) [\sigma(A^{(j)}) \bar{\sigma}(A m_e V) - \sigma(A) \bar{\sigma}(A^{(j)} m_e V)] = 0$; choose $\sigma(D) \neq 0$.

(F3) Flipping D_i gives $\sigma(D^{(i)}) \bar{\sigma}(A m_e V) = \sigma(A) \bar{\sigma}(V m_f D^{(i)})$. Multiply (1) by $\sigma(D^{(i)})$ and this by $\sigma(D)$ and subtract: $\sigma(A) [\sigma(D^{(i)}) \bar{\sigma}(V m_f D) - \sigma(D) \bar{\sigma}(V m_f D^{(i)})] = 0$; choose $\sigma(A) \neq 0$ (A is free here, since it does not occur on the right).

(F2) Flipping V_i gives $\sigma(D) \bar{\sigma}(A m_e V^{(i)}) = \sigma(A) \bar{\sigma}(V^{(i)} m_f D)$. Multiply (1) by $\bar{\sigma}(V^{(i)} m_f D)$ and this by $\bar{\sigma}(V m_f D)$ and subtract: $\sigma(D) [\bar{\sigma}(A m_e V) \bar{\sigma}(V^{(i)} m_f D) - \bar{\sigma}(A m_e V^{(i)}) \bar{\sigma}(V m_f D)] = 0$; choose $\sigma(D) \neq 0$. $\square$

8.2 Nonvanishing

Lemma 8.2 (nonvanishing). $\sigma(u) \neq 0$ for every u , and $\bar{\sigma}(y) \neq 0$ for every y .

Status. Proved below for $e \leq 2$; for $e \geq 3$ this is the one gap of the paper, and is discussed in §11. It is verified exhaustively for $(e, f) \in \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4)\}$ over finite fields (Check C5) and, at $e = 3$, for $(3, 4)$ and $(3, 5)$ by a Gröbner computation over every proper zero-pattern of W (Check C6). Everything after this point assumes it.

Proof for $e \leq 2$. For $e = 1$, σ is a single scalar, nonzero by hypothesis; and by Proposition 8.5 below $\bar{\sigma}$ is then a nonzero constant.

Let $e = 2$, so σ is defined on $\{0, 1\}$. Set $\delta = 1$ in (2) (π, π' both empty):

$$\sigma(0) \bar{\sigma}(0 \theta') = \sigma(1) \bar{\sigma}(1 \theta') \quad \text{for all } \theta' \in \{0, 1\}^{f-2}. \quad (*)$$

Suppose $\sigma(0) = 0$; then $\sigma(1) \neq 0$ and $(*)$ gives $\bar{\sigma}(1 \theta') = 0$ for all θ' . But by (F1) with $j = 1$, $\sigma(0) \bar{\sigma}(1 m V) = \sigma(1) \bar{\sigma}(0 m V)$, whence $\bar{\sigma}(0 m V) = 0$ for all m, V as well; so $\bar{\sigma} \equiv 0$, a contradiction. Symmetrically $\sigma(1) \neq 0$. Finally, by (F1) again, $\bar{\sigma}(y)$ and $\bar{\sigma}(y^{(1)})$ differ by the nonzero factor $\sigma(y_1^\vee)/\sigma(y_1)$, and by Lemma 8.4 below every $\bar{\sigma}(y)$ is a nonzero multiple of a single one. $\square$

8.3 Constancy of the flip ratios

Granting Lemma 8.2, define for $1 \leq j \leq e-1$ and $1 \leq k \leq f-1$

$$\theta_j(u) = \frac{\sigma(u | u_j = 1)}{\sigma(u | u_j = 0)}, \quad \bar{\theta}_k(y) = \frac{\bar{\sigma}(y | y_k = 1)}{\bar{\sigma}(y | y_k = 0)} \in F^\times,$$

functions of the remaining letters.

Lemma 8.3. Every θ_j and every $\bar{\theta}_k$ is a constant.

Proof. Step 1: $j \leq e-1$. By (F1), $\bar{\theta}_j(y) = \theta_j(y_{[1, e-1]})$; in particular $\bar{\theta}_j$ depends only on $y_{[1, e-1]}$. By (2) divided through,

$$\theta_\delta(\pi \pi') \cdot \bar{\theta}_{e-\delta}(\pi' \theta') = 1 \quad (1 \leq \delta \leq e-1), \quad (3)$$

where on the left $\pi\pi'$ denotes the word with π in positions $[1, \delta - 1]$ and π' in $[\delta + 1, e - 1]$, and on the right $\pi'\theta'$ the word with π' in $[1, e - \delta - 1]$ and θ' in $[e - \delta + 1, f - 1]$. The left factor does not involve θ' and the right factor does not involve π ; since their product is 1, each is independent of the variable the other lacks. Hence

- $\theta_\delta(u)$ depends only on $u_{[\delta+1, e-1]}$;
- $\bar{\theta}_k(y)$ depends only on $y_{[1, k-1]}$, for $k = e - \delta \in [1, e - 1]$.

Combining: for $1 \leq k \leq e - 1$, $\bar{\theta}_k$ depends only on $y_{[1, k-1]}$ and (being $\theta_k(y_{[1, e-1]})$) only on $y_{[k+1, e-1]}$. These index sets are disjoint, so $\bar{\theta}_k$ — and with it θ_k — is constant.

Step 2: $k = e$ and $k = f - e$. By (F0), $\bar{\theta}_e = \bar{\theta}_{f-e} = 1$.

Step 3: $e < k \leq f - 1$. Write $k = e + i$, $1 \leq i \leq f - e - 1$. Rearranging (F2),

$$\bar{\theta}_{e+i}(AmV) = \bar{\theta}_i(Vm'D),$$

where the left side is a function of V alone (it does not involve m', D) and the right side is a function of V alone (it does not involve A, m). So $\bar{\theta}_{e+i}$ and $\bar{\theta}_i$ are *the same function* of V , read in positions $[e + 1, f - 1]$ and $[1, f - e - 1]$ respectively. We induct on k : Steps 1–2 give constancy for $k \leq e$, and if $\bar{\theta}_i$ is constant then so is $\bar{\theta}_{e+i}$. Since every $k \in [e + 1, f - 1]$ is uniquely $e + i$ with $i \in [1, f - e - 1] < k$, all $\bar{\theta}_k$ are constant. $\square$

Lemma 8.4 (characters). *There are constants $\gamma_1, \dots, \gamma_{f-1} \in F^\times$ with*

$$\sigma(u) = \sigma(0^{e-1}) \prod_{j=1}^{e-1} \gamma_j^{u_j}, \quad \bar{\sigma}(y) = \bar{\sigma}(0^{f-1}) \prod_{k=1}^{f-1} \gamma_k^{y_k}.$$

Proof. Immediate from Lemma 8.3: $\bar{\theta}_k =: \gamma_k$ is a constant, so flipping letters one at a time from 0^{f-1} multiplies $\bar{\sigma}$ by the corresponding γ_k ; and $\theta_j = \bar{\theta}_j = \gamma_j$ for $j \leq e - 1$ by (F1). $\square$

8.4 Identification of γ

Proof of Theorem 5.2, necessity. Assume $W \not\equiv 0 \not\equiv \bar{W}$ and adopt Lemmas 8.2–8.4. Put $\alpha = W(0^{e-1}) = \sigma(0^{e-1})$ and $\beta = \bar{W}(0^{f-1})$. It remains to identify the constants γ_k .

- (F0) gives $\gamma_e = 1$ and $\gamma_{f-e} = 1$.
- (F2) gives $\gamma_{e+i} = \gamma_i$ for $1 \leq i \leq f - e - 1$: so $k \mapsto \gamma_k$ is e -periodic on $[1, f - 1]$. Extend it to a function γ on $\mathbb{Z}/e$; then $\gamma(0) = \gamma_e = 1$.
- (F3) gives $\gamma_{f-e+i} = \theta_i = \gamma_i$ for $1 \leq i \leq e - 1$: so γ is invariant under translation by $f - e \equiv f \pmod{e}$. The subgroup of $\mathbb{Z}/e$ generated by f is $d\mathbb{Z}/e\mathbb{Z}$ with $d = \gcd(e, f)$, so γ factors through $\mathbb{Z}/d$, still with $\gamma(0) = 1$.
- (3) gives $\gamma(\delta)\gamma(e - \delta) = 1$ for $1 \leq \delta \leq e - 1$; and $e \equiv 0 \pmod{d}$, so this is $\gamma(\delta)\gamma(-\delta) = 1$ on $\mathbb{Z}/d$ (the case $\delta \equiv 0$ being $\gamma(0)^2 = 1$, which holds).

Thus $\sigma = \alpha\sigma_e^\gamma$ and $\bar{\sigma} = \beta\sigma_f^\gamma$, i.e. $W = \alpha W_e^\gamma$ and $\bar{W} = \beta W_f^\gamma$, which is case (c). Conversely (a), (b), (c) all give $[R_e^W, R_f^{\bar{W}}] = 0$, by inspection and Proposition 7.1. $\square$

Proposition 8.5 (the case $e = 1$, without the two-bead sector). *Let $e = 1 < f$ and $W(\emptyset) = \alpha \neq 0$. Then $[R_1^W, R_f^{\bar{W}}] = 0$ iff $\bar{\sigma}$ is constant, i.e. $\bar{W}(y) = \beta(-1)^{|y|}$.*

Proof. Here $A = D = \emptyset$ and (1) reads $\bar{\sigma}(m_e V) = \bar{\sigma}(V m_f)$ for all $m_e, m_f \in \{0, 1\}$ and $V \in \{0, 1\}^{f-2}$. Taking $m_e = 0$: $\bar{\sigma}(0V) = \bar{\sigma}(V m_f)$ for both m_f , so $\bar{\sigma}(y)$ is unchanged when the first letter is set to 0 and the word is cyclically shifted left. Iterating $f - 1$ times turns any y into 0^{f-1} . Hence $\bar{\sigma}$ is constant. This matches Theorem 5.2 because $d = \gcd(1, f) = 1$. $\square$

9 What the extra freedom is: the d -quotient, and gauge

9.1 The special points: Murnaghan–Nakayama one level down

Proposition 9.1. *For each divisor $d' \mid d$, the function $\gamma(\delta) = (-1)^{[d'\delta]}$ satisfies the conditions of Theorem 5.2(c), and the corresponding weight is*

$$W(u) = \alpha (-1)^{\#\{i \in d'\mathbb{Z} \cap [1, e-1] : u_i=1\}}.$$

Under the d' -quotient isomorphism, which splits the Maya set into the d' runners $M^{(r)} = \{k : d'k + r \in M\}$, this weight counts exactly the beads jumped on the mover's own runner. So R_e^W becomes $\alpha \sum_r (p_{e/d'} \cdot)_r$ and R_f^W becomes $\beta \sum_r (p_{f/d'} \cdot)_r$, which commute because $p_{e/d'}$ and $p_{f/d'}$ do.

Proof. $\gamma(0) = 1$ and $\gamma(\delta)\gamma(-\delta) = (\pm 1)^2 = 1$ since $d' \mid \delta$ iff $d' \mid -\delta$. For the weight: $\sigma^\gamma(u) = \prod_j \gamma(j)^{u_j} = (-1)^{\#\{i: d'i, u_i=1\}}$, and $W = (-1)^{|u|}\sigma$ multiplies this by $(-1)^{|u|}$, leaving $(-1)^{\#\{i: d'i, u_i=1\}}$. The sites $b + i$ with $d' \mid i$ are precisely those on the runner of b . The rest is Proposition 7.1. $\square$

So $d' = 1$ is Murnaghan–Nakayama itself, and $d' = e$ (when $e \mid f$) is the *unsigned* hopping operator $W \equiv \alpha$. For $(e, f) = (2, 4)$ these are the only two points of the locus, and both are visible in the raw equations (§10, Check E2).

9.2 The torus directions are a diagonal gauge

Proposition 9.2 (an explicit gauge). *For a Maya set M and $\delta \in \mathbb{Z}/d$, $\delta \neq 0$, put*

$$\Lambda_\delta(M) := \#\{(i, j) : i < j, i \notin M, j \in M, j - i \equiv \delta \pmod{d}\} < \infty.$$

(Finiteness: $i \notin M$ forces i bounded below and $j \in M$ forces j bounded above.) Let $x_1, \dots, x_{d-1} \in F^\times$ and let D be the diagonal operator $|M\rangle \mapsto (\prod_\delta x_\delta^{\Lambda_\delta(M)})|M\rangle$. Then for every e with $d \mid e$,

$$D R_e^{W^{\text{MN}}} D^{-1} = \left(\prod_\delta x_\delta^{e/d} \right) R_e^{W^\gamma}, \quad \gamma(\delta) = \frac{x_\delta}{x_{-\delta}},$$

where $W^{\text{MN}}(u) = (-1)^{|u|}$. Consequently every γ of the form $\delta \mapsto x_\delta/x_{-\delta}$ — that is, every γ in the connected component of the identity of the group of Corollary 5.4 — is gauge.

Proof. Let $M' = (M \setminus \{b\}) \cup \{b + e\}$ and $A_\delta = \#\{i \in M \cap (b, b + e) : i - b \equiv \delta\}$. Counting the pairs gained and lost, and using $d \mid e$ throughout so that $b + e \equiv b$: the pairs with $j = b$ are lost and those with $j = b + e$ gained, contributing $\#\{i \in (b, b + e) : i \notin M, i - b \equiv -\delta\}$; the pairs with $i = b + e$ are lost and those with $i = b$ gained, contributing $\#\{j \in (b, b + e) : j \in M, j - b \equiv \delta\}$. The interval $(b, b + e)$ contains exactly e/d sites in each nonzero residue class mod d , so the first count is $e/d - A_{-\delta}$. Hence $\Lambda_\delta(M') - \Lambda_\delta(M) = e/d + A_\delta - A_{-\delta}$ and

$$\frac{D_{M'}}{D_M} = \left(\prod_\delta x_\delta^{e/d} \right) \prod_\delta \left(\frac{x_\delta}{x_{-\delta}} \right)^{A_\delta} = \left(\prod_\delta x_\delta^{e/d} \right) \sigma^\gamma(u_M(b, e)),$$

which is the claim after multiplying by $W^{\text{MN}} = (-1)^{|u|}$. $\square$

Remark 9.3 (the residual, and what remains of it). The map $(x_\delta) \mapsto (\delta \mapsto x_\delta/x_{-\delta})$ surjects onto $\{\gamma : \gamma(\delta)\gamma(-\delta) = 1\}$ *except* at a fixed point $\delta = -\delta$, where it can only produce $\gamma(\delta) = 1$ while the theorem allows $\gamma(\delta) = -1$. So Proposition 9.2 accounts for the whole locus when d is odd, and for an index-2 subgroup when d is even. The missing $\mathbb{Z}/2$ at $\delta = d/2$ is nevertheless *also* a gauge, by a different (cross-runner inversion) statistic: this was verified computationally — a spanning-tree consistency test for the cocycle $M \rightarrow M' \mapsto W^\gamma/W^{\text{MN}}$ found 0 violations over 69 686 and 81 119 edges on 15 000 states for $(3, 6)$ and $(2, 4)$, while non-commuting controls produced 25 304 and 23 381 violations (§10, Check C7). I have not written out the statistic in general, and mark it as a gap (§11).

9.3 Transpose-equivariance is exactly the two-bead condition

Proposition 9.4. *Let $\sigma(u) = \prod_j \gamma(j \bmod d)^{u_j}$ with $\gamma(0) = 1$ and $d \mid e$. Then*

$$\sigma \circ T = \kappa \sigma \quad \text{for a constant } \kappa \quad \Longleftrightarrow \quad \gamma(\delta)\gamma(-\delta) = 1 \quad \forall \delta,$$

with T the transposition of Lemma 1.6; and then $\kappa = \prod_{j=1}^{e-1} \gamma(j \bmod d)$ and $W \circ T = (-1)^{e-1} \kappa W$.

Proof. $\sigma(u^T) = \prod_j \gamma(j)^{1-u_{e-j}} = \kappa \prod_{j'} \gamma(e-j')^{-u_{j'}} = \kappa \prod_{j'} \gamma(-j')^{-u_{j'}}$, using $d \mid e$. This equals $\kappa \sigma(u)$ for all u iff $\gamma(-j')^{-1} = \gamma(j')$ for every j' , i.e. iff $\gamma(j')\gamma(-j') = 1$. For W , use $|u^T| = e - 1 - |u|$. $\square$

This is the sharpest way to say what the answer is. The one-bead sector says the weight must be a d -periodic character of the occupancy word; the two-bead sector says it must be equivariant under transposing the ribbon. Murnaghan–Nakayama is the unique weight that is both when $d = 1$, because $\mathbb{Z}/1$ has only the trivial character.

Remark 9.5 (which sector does what — and the surprise). In [1] the one-bead sector was the strong one: it forced the anchor outright, while the two-bead sector only forced multiplicativity. Here the roles are the other way round. The one-bead sector is *blind* to γ beyond $\gamma(0) = 1$ — it imposes no condition at all on the free parameters — and the entire cut is made by the two-bead sector. That is why the phenomenon is invisible at $\min(e, f) = 1$, where there is no two-bead sector: there $d = 1$ anyway and the one-bead sector suffices.

10 Computational verification

All code is in `proofs/code/2026-09-15-c1-Q149/`. The engine takes an *opaque weight callable* indexed by occupancy words; it never forms $(-1)^{|u|}$ or $t^{|u|}$, so it derives the locus rather than testing a guess.

V1 — two independent ribbon enumerations, with shape. `ribbons_young` enumerates border strips by brute force over Young diagrams and reads the row-length composition directly; `ribbons_maya` performs the bead move and reads the occupancy word. They agree, via Lemma 1.3, on all 1375 ribbons with $|\lambda| \leq 8$, $e \leq 5$. This check *failed* before the reversal of Lemma 1.3 was inserted.

V2 — transposition. Reverse-and-complement on words matches conjugation of Young diagrams on 1500 ribbons, $e \leq 7$, $|\lambda| \leq 7$ (1500/1500). Plain reversal matches on 75/765.

E1, E2 — the raw equations. The commutator is applied to states and the coefficients collected as polynomials in the opaque symbols. Two state enumerations are used and agree exactly on the resulting equation set: all charge-0 Maya sets in an $L \times L$ box ($\binom{2L}{L}$ states), and all window configurations $M = \mathbb{Z}_{<0} \cup T$, $T \subseteq [0, L_w)$. The equation set stabilises at $L_w = e + f + 1$.

(e, f)	states	elements	distinct nonzero equations
(1, 3)	252	4660	5
(2, 3)	924	24892	17
(2, 4)	3432	125712	34

At (1, 3) the five equations are $\alpha \cdot (\bar{W}_{00} + \bar{W}_{01})$, $\alpha \cdot (\bar{W}_{01} + \bar{W}_{11})$, $\alpha \cdot (\bar{W}_{10} - \bar{W}_{01})$, $\alpha \cdot (\bar{W}_{00} + \bar{W}_{10})$, $\alpha \cdot (\bar{W}_{10} + \bar{W}_{11})$ — the third being $\bar{W}_{(2,1)} = \bar{W}_{(1,2)}$ (Corollary 5.5).

C1 — calibration to [1]. Substituting height weights $W(u) = w_{|u|}$ into the shape engine returns exactly the Q147 relation $w_z \bar{w}_{u+v+1} + w_u \bar{w}_{v+z} = 0$: all instances are recovered, for $(e, f) \in \{(1, 3), (2, 3), (1, 4), (2, 4), (2, 5), (3, 4)\}$ (2/2, 4/4, 3/3, 8/8, 12/12, 9/9). Without this the generalised engine would mean nothing.

C2 — realisability. All 2^{e-1} occupancy words occur among partitions of size ≤ 14 , for $e \leq 6$. No parameter is free for a trivial reason.

C3 — the new direction is measured. The residual after the forced relations contains a genuinely shape-dependent parameter. At (3, 6) the symbolic family $\gamma = (1, t, t^{-1})$ annihilates the commutator *identically in t* (0 residuals); the negative control $\gamma = (1, t, t)$, which violates $\gamma(\delta)\gamma(-\delta) = 1$, leaves 10 residuals, all divisible by $t^2 - 1$. The control moves the prediction.

C4 — $e = f$ manufactures nothing. For $e = f \in \{1, 2, 3, 4\}$ the solver returns 0 equations from 22, 220, 1608, 10144 matrix elements respectively.

C5 — exhaustive classification over finite fields. For small (e, f) every tuple $(W, \bar{W}) \in \mathbb{F}_p^{2^{e-1}} \times \mathbb{F}_p^{2^{f-1}}$ was tested against the raw equations and compared with Theorem 5.2. The two sets are *equal*, with no extra and no missing solutions:

(e, f)	p	solutions	predicted
(1, 3)	5	645	$5^4 + 5 - 1 + 4^2 \cdot 1$
(2, 3)	5	665	$5^4 + 5^2 - 1 + 4^2 \cdot 1$
(1, 4)	3	6567	$3^8 + 3 - 1 + 2^2 \cdot 1$
(2, 4)	3	6577	$3^8 + 3^2 - 1 + 2^2 \cdot 2$

The last row is the decisive one: $d = \gcd(2, 4) = 2$, and the factor 2 in $2^2 \cdot 2$ is the extra branch of Corollary 5.4. This check covers the degenerate strata as well, and so confirms Lemma 8.2 for these pairs.

C6 — nonvanishing at $e = 3$. For (3, 4) and (3, 5), each of the $2^4 - 2$ proper nonempty zero-patterns of W was substituted and the resulting ideal saturated; in every case $\bar{W} \equiv 0$ is forced. So no solution has W vanishing partially.

C7 — the gauge test. The edge function $\eta(M \rightarrow M') = W^\gamma(u)/W^{\text{MN}}(u)$ on the graph of e - and f -moves was tested for exactness by assigning potentials along a breadth-first spanning tree from the vacuum and checking every non-tree edge. Commuting families: 0 violations ((3, 6), 69 686 edges; (2, 4), 81 119 edges; 15 000 states each). Non-commuting controls: 25 304 and 23 381 violations, with ratios $t^{\pm 2k}$. The test discriminates.

C8 — the family against the full commutator. For every $d' \mid \gcd(e, f)$ the weight of Proposition 9.1 annihilates the commutator on the complete charge-0 enumeration: (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (2, 5), (2, 6), up to 48 620 states and 2 767 908 matrix elements, 0 nonzero residuals throughout. Negative controls with $d' \mid e$ but $d' \nmid f$ give 2 nonzero residuals each.

Count audit. The one-bead equations are indexed by $(A, m_e, V, m_f, D) \in \{0, 1\}^{e-1} \times \{0, 1\} \times \{0, 1\}^{f-e-1} \times \{0, 1\} \times \{0, 1\}^{e-1}$, i.e. $2^{e-1} \cdot 2 \cdot 2^{f-e-1} \cdot 2 \cdot 2^{e-1} = 2^{e+f-1}$ window words X — one per configuration of the $g - 1$ free sites of $(b, b + g)$, as it must be. The two-bead equations are indexed by $\delta \in [1, e - 1]$ together with $(\pi, \pi', \theta') \in \{0, 1\}^{\delta-1} \times \{0, 1\}^{e-\delta-1} \times \{0, 1\}^{\delta+f-e-1}$, i.e. $\sum_{\delta=1}^{e-1} 2^{f-2} = (e-1)2^{f-2}$; this is 0 when $e = 1$, as it must be. Neither total is quoted bare: at (1, 3) the $2^3 = 8$ window words give 5 distinct nonzero equations after the m -independent ones collapse, and $(e-1)2^{f-2} = 0$ two-bead equations, matching the table in E1.

11 What is and is not proved

Proved

1. Lemmas 1.3 and 1.6 (the dictionary and the correct transposition rule), with computational confirmation and a correction to the brief (Remark 1.7).
2. Lemmas 4.2 and 4.3: both sectors' matrix elements for free shape weights, and the fact that both are *sign-free* in σ -coordinates. The Murnaghan–Nakayama sign is a change of coordinates, not a feature of the locus.
3. Theorem 5.2, sufficiency (Proposition 7.1): complete and unconditional.
4. Theorem 5.2, necessity: complete *given* Lemma 8.2. The chain is Lemma 8.1 $\rightarrow$ Lemma 8.3 $\rightarrow$ Lemma 8.4 $\rightarrow$ identification of γ .
5. Lemma 8.2 for $e \leq 2$; Proposition 8.5 for $e = 1$ without any two-bead input.
6. Corollary 5.3: for $\gcd(e, f) = 1$, shape dependence is forced away and the Q147 theorem holds verbatim on the 2^{e-1} -parameter family. In particular the answer to the brief's headline question is **yes at** (1, 3): the cross-rank commutator forces $\bar{W}_{(2,1)} = \bar{W}_{(1,2)}$.
7. Proposition 9.1 (the d' -quotient points) and Proposition 9.2 (an explicit diagonal gauge realising the identity component of the residual torus).
8. Proposition 9.4: the two-bead condition is exactly transpose-equivariance.

Not proved

(G1) **Lemma 8.2 for $e \geq 3$.** I need: if $[R_e^W, R_f^{\bar{W}}] = 0$ with $W \neq 0 \neq \bar{W}$, then no value of W or $\bar{W}$ vanishes. What I have: from (1) with $\sigma(A_0) = 1$ one gets $\bar{\sigma}(AmV) = \sigma(A)\rho(V)$ and

$\bar{\sigma}(VmD) = \rho(V)\sigma(D)$ with $\rho(V) := \bar{\sigma}(A_00V)$, so the zero set of $\bar{\sigma}$ is controlled by those of σ and ρ ; and from (2) the zero set of σ is closed under flipping position δ *unless* $\bar{\sigma}$ vanishes on a whole cylinder, which in turn forces σ to vanish on a cylinder. The bookkeeping closes for $e = 2$ (where the cylinders are single words) and I could not close it for $e \geq 3$; the obstruction is that the cylinder produced has length $e - \delta$, not $e - 1$, so the induction does not terminate in one step. Verified computationally for $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4)$ exhaustively and for $(3, 4), (3, 5)$ by saturation (Checks C5, C6). *This is the only gap in the classification.*

- (G2) **The $\mathbb{Z}/2$ at $\delta = d/2$ is gauge.** Proposition 9.2 produces the identity component only. For $d = 2$ I can exhibit the missing gauge by hand — the cross-runner inversion number $N(M) = \#\{(p, q) \in M^{(0)} \times M^{(1)} : q < p\}$, suitably regularised against the vacuum, has increment $[\beta \in M^{(1)}]$ under a move on runner 0 and $-[\beta + 1 \in M^{(0)}]$ under a move on runner 1, which is exactly the required sign — but I have not written the regularisation carefully nor generalised it to $d > 2$. Check C7 is strong evidence (0 violations on $\sim 1.5 \times 10^5$ edges with a control that fires $\sim 5 \times 10^4$ times), not a proof.
- (G3) **Shared weights.** Theorem 5.2 is stated for independent $W, \bar{W}$. The shared case (W and $\bar{W}$ restrictions of one function) is a further intersection I have not computed; note that the γ is already shared in (c), so the only extra condition is $\alpha = \beta$ up to the normalisation of Convention 5.1.
- (G4) **More than two ranks.** Whether requiring $[R_e^{W_e}, R_f^{W_f}] = 0$ for *all* pairs $e < f$ from a set forces $d = \gcd$ over the whole set is not addressed; the expected answer, γ periodic mod $\gcd$ of the whole set, would follow by intersecting the loci, but I have not checked that the intersection is the naive one.

The headline, stated honestly

The brief offered three outcomes. The answer is **(B) with a (C) qualifier**, and the qualifier does not weaken it:

When $\gcd(e, f) = 1$ — in particular for the brief’s decisive test $(e, f) = (1, 3)$ — the Murnaghan–Nakayama sign is the only dressing of the ribbon under which different sizes commute, and this is now known for the full 2^{e-1} -parameter shape family, not just for height weights. When $\gcd(e, f) = d > 1$ the locus is strictly larger — of dimension $\lfloor (d-1)/2 \rfloor$ — but every extra point is Murnaghan–Nakayama transported through a d' -quotient ($d' \mid d$) or twisted by a diagonal gauge. What is *genuinely* pinned down is a statement the height-only theorem could not see: **the weight must be a $\gcd(e, f)$ -periodic character of the occupancy word that is equivariant under transposing the ribbon**, and the two conditions are contributed by the one-bead and two-bead sectors respectively.

References

- [1] Clio, *Free height weights force Murnaghan–Nakayama* (Q147), `proofs/2026-09-11-c2-Q147-multi-t-bracket.tex`, 11 September 2026.
- [2] Clio, *Ribbon commutators on Maya diagrams* (Q92), `proofs/`, September 2026. Used for: the sector lemma, the determination of the moved pair when $e \neq f$, and the emptiness of the two-bead sector at $\min(e, f) = 1$.